



UNIVERSITY
Of KIGALI

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ASSIGNMENT 1	
GRADUATE SCHOOL	
PROGRAMME: Master of Science in Information Technology	
MODULE CODE AND TITLE: MIT91207: Machine Learning	LEVEL AND TRIMESTER: LEVEL 9, TRIMESTER 4
SUBMISSION DATE: September 17, 2026, at 23:59	SESSION: WEEKEND
LECTURER: DR. GUSTAVE UDAHEMUKA	TOTAL MARKS: 102 TO BE SCALED TO 15 MARKS

QUESTION 1 (12 Marks)

A university intends to develop a data-driven early-warning system that can identify students who may be at risk of poor academic outcomes. The university has accumulated information including attendance records, previous grades, assessment scores, participation in extracurricular activities, engagement with the learning management system, and other student-related attributes. University management wants the system to support academic advisory teams in deciding which students may require additional intervention.

You have been appointed as a machine-learning consultant to assess the feasibility of the proposed system.

Required:

a) Formulate the university's problem as a machine-learning problem and determine whether the primary task should be treated as classification, regression, or potentially both. Justify your decision in terms of the target variable and the type of decision the university wants to make.

(5 Marks)

b) Design two alternative modelling approaches that could be used for the problem: one classification-based and one regression-based. For each approach, identify an appropriate algorithm and explain how its predictions could support an academic intervention decision. **(5 Marks)**

c) The university is concerned that a highly accurate model may still make inappropriate intervention decisions. Evaluate two factors other than predictive accuracy that should be considered before deploying the system and explain how they could affect the reliability or acceptability of the solution. **(2 Marks)**

QUESTION 2 (13 Marks)

An online learning platform is redesigning its course-recommendation system. The platform has historical records containing learner IDs, course titles, course categories, ratings, completion status, timestamps, and the number of courses previously taken by each learner. An initial inspection reveals missing ratings, duplicated records, inconsistent category names, and numerical variables recorded on substantially different scales.

The development team intends to use machine-learning techniques to recommend courses that learners are likely to value.

Required:

a) Design a preprocessing pipeline for the dataset. Your solution should determine how the identified data-quality problems should be addressed and explain how the preprocessing decisions could influence the subsequent machine-learning model. **(5 Marks)**

b) Using Python and pandas, implement a preprocessing procedure that demonstrates at least three appropriate data-cleaning or feature-engineering operations on a sample version of the dataset. **(5 Marks)**

c) Create at least two meaningful features that could improve the representation of learner behaviour. Justify why each feature may be useful for the recommendation problem. **(2 Marks)**

d) Explain how you would ensure that information from the future does not inadvertently enter the training data when timestamped historical records are used. **(1 Mark)**

QUESTION 3 (13 Marks)

A financial institution is developing a machine-learning system to identify customers who are likely to default on loans. The dataset contains income, employment information, previous credit history, loan amount, repayment history, and other customer attributes. During an initial audit, the institution discovers that some records contain missing income values, incomplete employment information, and missing credit-history attributes.

The management team proposes simply replacing all missing numerical values with zero before training the model because this would allow the machine-learning algorithm to process the complete dataset.

Required:

- a) Critically evaluate the proposed strategy of replacing missing values with zero. Use the characteristics of financial data to explain circumstances in which this approach could introduce misleading information into the model. **(5 Marks)**
- b) Develop a more appropriate missing data strategy for the institution. Your strategy should distinguish between different types of variables and consider the possible reasons why information is missing. **(3 Marks)**
- c) Design a Python function that can be incorporated into a data-preprocessing pipeline to identify and handle missing values using an appropriate strategy. Explain the assumptions underlying your implementation. **(3 Marks)**
- d) Evaluate how inappropriate treatment of missing data could affect both model performance and decisions made about individual loan applicants. **(2 Marks)**

QUESTION 4 (12 Marks)

An organization has collected a dataset for predicting whether customers are likely to renew a subscription. The dataset contains numerical attributes such as age, monthly expenditure, number of previous transactions, and account duration, together with categorical attributes such as subscription type, geographical region, payment method, and customer status.

Before model training, the data-science team must transform these heterogeneous variables into a representation suitable for a machine-learning algorithm.

Required:

- a) Design a feature-engineering pipeline for the dataset. Your proposal should specify how you would treat categorical and numerical variables and justify the choice of encoding and scaling techniques. **(3 Marks)**
- b) Evaluate the consequences of using one-hot encoding, ordinal encoding, and inappropriate numerical encoding for categorical variables. Identify circumstances in which each approach would or would not be appropriate. **(4 Marks)**
- c) Develop a Python/pandas implementation that transforms a representative dataset containing both categorical and numerical attributes into a model-ready feature representation. **(3 Marks)**
- d) The organization plans to deploy the model on newly arriving customer records. Explain how the feature-engineering process should be designed to ensure that the same transformations are consistently applied during deployment. **(2 Marks)**

QUESTION 5 (13 Marks)

A health-technology company intends to develop a predictive system for identifying patients who may be readmitted to hospital within a specified period after discharge. The available historical records contain demographic information, previous admissions, diagnoses, treatment information, length of stay, and other clinical attributes.

The company has asked an IT/ML team to develop a prototype that can later be integrated into its information system. The team must therefore consider not only model training but also data preparation, validation, evaluation, and eventual deployment.

Required:

- a) Design an end-to-end machine-learning workflow for the proposed system, beginning with data acquisition and ending with model evaluation and deployment considerations. Justify the major decisions in your workflow. **(4 Marks)**
- b) Propose an appropriate strategy for dividing the available data into training, validation, and testing subsets. Explain why a careless partitioning strategy could produce an overly optimistic estimate of model performance. **(3 Marks)**
- c) The development team obtains a model with high overall accuracy but discovers that its performance differs considerably between patient groups. Evaluate the situation and propose additional evaluation measures or procedures that should be considered before deployment. **(3 Marks)**
- d) Demonstrate in Python how an appropriate train-test or train-validation-test partition could be implemented. **(3 Marks)**

QUESTION 6 (13 Marks)

A government is developing a data-driven approach to understand the factors associated with differences in crime rates across geographical areas. As part of the investigation, an IT analytics team has been provided with the US Crime Dataset available on Kaggle, which contains crime rates together with socioeconomic, demographic, law-enforcement, and other characteristics for 47 US states.

The dataset available at <https://www.kaggle.com/datasets/shilpagopal/us-crime-dataset> defines Crime as the number of offenses per 100,000 population.

You have been appointed as a data analyst to investigate which available factors are most useful for predicting crime rates. The government emphasizes that the analysis should support evidence-based decision-making and should not confuse statistical association with causation.

Required:

- a) Using the provided dataset, develop a feature-selection strategy for predicting Crime. Explain how you would determine which variables should be retained, removed, transformed, or combined. **(3 Marks)**
- b) Using Python, perform exploratory analysis and construct a correlation matrix for the numerical variables. Use the results to identify potentially useful relationships and discuss why correlation alone should not be treated as sufficient evidence for selecting predictive features. **(6 Marks)**
- c) Identify at least one potential problem caused by highly correlated predictor variables and propose an appropriate strategy for dealing with it. **(2 Marks)**
- d) Based on your analysis, formulate recommendations to the government regarding the use of the selected variables for crime-resource planning. Clearly distinguish between statistical association and causal interpretation. **(2 Marks)**

QUESTION 7 (14 Marks)

YEGO, a ride-hailing company operating in Rwanda, wants to improve the accuracy of its estimated time of arrival (ETA) for its rides. Historical data contain information such as journey distance, traffic conditions, time of day, weather conditions, road characteristics, and previous journey times. The company is considering a linear regression model but is concerned that some predictors may be highly correlated and that some variables may contribute little useful information.

The data-science team proposes comparing Ordinary Least Squares Linear Regression, Ridge Regression, and Lasso Regression before selecting a model for deployment.

Required:

- a) Design an experimental procedure for comparing the three regression approaches. Your procedure should specify how the data will be prepared, how the models will be trained, and which evaluation measures should be used. **(3 Marks)**
- b) Using Python and the provided synthetic dataset '*synthetic_ETA.csv*', implement Linear Regression, Ridge Regression, and Lasso Regression and compare their predictive performance. **(6 Marks)**
- c) Analyse the results and determine which model would be the most appropriate for the ride-hailing company. Your decision should consider both predictive performance and model complexity rather than relying on a single numerical metric. **(2 Marks)**
- d) The company wants the final system to operate reliably on previously unseen journeys. Evaluate at least two measures that should be taken to reduce the risk of overfitting and obtain a more reliable estimate of real-world performance. **(3 Marks)**

QUESTION 8 (12 Marks)

A technology company is considering launching a new digital product. Before committing significant financial resources, the marketing team conducts a customer survey covering age group, occupation, previous use of similar products, preferred features, expected price range, satisfaction levels, and willingness to purchase.

The management team has provided the raw survey dataset to an IT analytics team and expects evidence-based recommendations rather than a simple summary of the responses.

Required:

- a) Develop an exploratory data-analysis strategy for extracting decision-relevant information from the survey data. Your approach should consider data quality, distributions, relationships between variables, and possible subgroups of customers. **(3 Marks)**
- b) Using Python, generate appropriate visualizations to investigate at least three important relationships or patterns in the survey data. Select visualizations based on the characteristics of the variables rather than using charts indiscriminately. **(4 Marks)**
- c) Interpret the findings and identify at least two insights that could influence the product launch decision. **(3 Marks)**
- d) Evaluate one limitation of drawing business conclusions solely from the survey results and propose additional evidence that the company should consider before making the final decision. **(2 Marks)**

END
